

Bachelor of Technology — Civil Engineering

Table 1: Civil Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	17
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Civil Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
2	CE-203	Construction Materials	3-0-0-3	3
3	CE-203P / CE-354P	Building (Constructions) Materials Lab	0-0-2-1	1
4	CE-252	Geology and Geomorphology	2-0-2-3	3
5	CE-301 / CE-310	Strength of Materials and Structures	3-0-1-3	3
6	CE-301P / CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
7	CE-201	Surveying Traditional and Digital	2-0-4-4	4
8	CE-251	Hydraulics Engineering	3-0-0-3	3
9	CE-302 / CE-311	Geotechnical Engineering I	3-0-0-3	3
10	CE-302P / CE-311P	Geotechnical Engineering Lab		1
11	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
12	CE-404	Analysis of Structures	3-0-0-3	3
13	CE-303	Water Resources Engineering	3-0-0-3	3
14	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
15	CE-352	Transportation Engineering	3-0-0-3	3
16	CE-352P	Transportation Engineering Lab	0-0-2-1	1
17	CE-353P	Civil Engineering Drawing	0-0-2-1	1
18	CE-402	Geotechnical Engineering II	2-1-0-3	3
19	CE-305P	Environmental Engineering Lab	0-0-2-1	1
20	CE-356	Reverse Engineering	1-0-0-1	1
21	CE-401	Design of Steel Structures	2-1-0-3	3
22	CE-403	Water and Wastewater Engineering	3-0-0-3	3
Total				49

Table 3: B. Tech Civil Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Civil Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Civil Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
4	DC	CE-203	Construction Materials	3-0-0-3	3
5	DC	CE-203P / CE-354P	Building (Constructions) Materials Lab	0-0-2-1	1
6	DC	CE-252	Geology and Geomorphology	2-0-2-3	3
7	DC	CE-301 / CE-310	Strength of Materials and Structures	3-0-1-3	3
8	DC	CE-301P / CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Civil Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-201	Surveying Traditional and Digital	2-0-4-4	4
2	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
3	DC	CE-302 / CE-311	Geotechnical Engineering I	3-0-0-3	3
4	DC	CE-302P / CE-311P	Geotechnical Engineering Lab		1
5	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
6	DC	CE-404	Analysis of Structures	3-0-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Civil Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
3	DC	CE-352	Transportation Engineering	3-0-0-3	3
4	DC	CE-352P	Transportation Engineering Lab	0-0-2-1	1
5	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
6	DC	CE-402	Geotechnical Engineering II	2-1-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Civil Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-305P	Environmental Engineering Lab	0-0-2-1	1
2	DC	CE-356	Reverse Engineering	1-0-0-1	1
3	DC	CE-401	Design of Steel Structures	2-1-0-3	3
4	DC	CE-403	Water and Wastewater Engineering	3-0-0-3	3
5	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Civil Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Civil Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CE-010	Internship		2
2	MTP	CE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Beyond the core courses above, students choose 17 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).

Table 11: Civil Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CD-591	Special Topics in Disaster Resilient Hill Roads	1
2	CE-304	Hydraulics Engineering Lab	3
3	CE-305	Environment and Earth Science Lab	3
4	CE-306	Environmental Engineering Lab	3
5	CE-353	Civil Engineering Drawing	3
6	CE-354	Building and Pavement Materials Laboratory	3
7	CE-355	Hydrology	3
8	CE-405	Water and Wastewater Engineering	3
9	CE-451	Irrigation Engineering and Hydraulic Structures	3
10	CE-452	Open Channel Flow	3
11	CE-501	Remote Sensing	3
12	CE-502	Groundwater Flow and Contaminant Transport	3
13	CE-503	Fundamentals of Project Management	3
14	CE-504	Slope Stability and Retaining Structures	3
15	CE-505	Engineering of Ground Modification	3
16	CE-506	Analysis of Indeterminate Structures	3
17	CE-507	Advanced Concrete Science	3
18	CE-508	Photogeology and Photogrammetry	3
19	CE-509	Bridge Engineering	3
20	CE-510	Modelling and Simulation in Water Resources Engineering	3
21	CE-511	Structural Dynamics with Application to Earthquake Engineering	3
22	CE-512	Advanced Soil Mechanics	3
23	CE-513	Advanced Foundation Engineering	3
24	CE-514	Rock Mechanics	3
25	CE-515	Environmental Impact Assessment	3
26	CE-516	Uncertainty Analysis in Civil Engineering	3
27	CE-517	Hydroinformatics	4
28	CE-518	Structural Reliability and Risk Assessment	3
29	CE-519	Chemistry of Natural Waters	3
30	CE-520	Environmental Reaction Modeling	3
31	CE-521	Ecology and Environment Microbiology	4
32	CE-522	Matrix Methods for Structural Analysis	3
33	CE-523	Building Science	3
34	CE-524	Applied Hydroclimatology	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
35	CE-524P	Computational Hydroclimatology Lab	1
36	CE-526	Critical Zone Science	3
37	CE-551	Geosynthetics and their applications	3
38	CE-552	Concrete Technology	3
39	CE-554	Prestressed Concrete Structures	3
40	CE-555	Advanced Design of Structures	3
41	CE-556	Structural Engineering Laboratory	3
42	CE-556P	Structural Engineering laboratory	2
43	CE-557	Solid Mechanics in Structural Engineering	3
44	CE-558	Air pollution and its mitigation	3
45	CE-559	Biological Wastewater Treatment	3
46	CE-560	Soil Dynamics	3
47	CE-561	The science of climate change	3
48	CE-562	Numerical Methods for Engineering Computation	3
49	CE-563	Finite Element Methods in Engineering	4
50	CE-564	Fundamentals of Fracture Mechanics	3
51	CE-567	Highway Safety and Modeling	3
52	CE-568P	Experimental Structural Dynamics	1
53	CE-586	Mini Project	3
54	CE-586P	Mini Project	3
55	CE-587	Industrial / Academic Internship	3
56	CE-591	Special Topics in Civil and Mechanical Engineering	3
57	CE-600	Research Methodology for Civil Engineering	1
58	CE-601	Geo-Informatics	3
59	CE-602	Blast Engineering	3
60	CE-604	Theory of Plasticity	3
61	CE-605	Engineering Seismology and Seismic Hazard Assessment	3
62	CE-606	Constitutive Modeling of Frictional Material	3
63	CE-610	Analysis and Design for Earthquake Resistant Structures	3
64	CE-611	Structural Health Monitoring	3
65	CE-612	Theory of Plates and Shells	3
66	CE-613	Mechanics of Unsaturated Soils	3
67	CE-617	Mechanics of Composite Materials	3
68	CE-688	Post Graduate Project - I	3
69	CE-689	Post Graduate Project - II	3
<i>Compulsory project components — counted as DE, not selectable</i>			
70	CE-498P	Major Technical Project - I (compulsory; counted as DE)	4

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
71	CE-499P	Major Technical Project - II (compulsory; counted as DE)	4